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Abstract

SME online retailers need transparent 60-day repurchase risk scores backed by concrete invoice evidence — not black-box predictions

GraphRAG Foundation & Current Strength

Domain Online retail (UCI Online Retail II)
Goal Interpretable 60-day repurchase risk with full invoice provenance

Nodes
Customer · Invoice · InvoiceLine · Product · Community · FeatureVector

Key Relations
PURCHASED · BELONGS_TO · PLACED · CONTAINS · SHARES_COMMUNITY

Community features lift PR-AUC
+0.08 (0.31 → 0.39)

Current Limitations

- Template routing fails on novel phrasings
- Sample graph too small for realistic ranking
- Limited multimodal community explanations
- No interactive community exploration

What Communities Mean & How We Detect Them

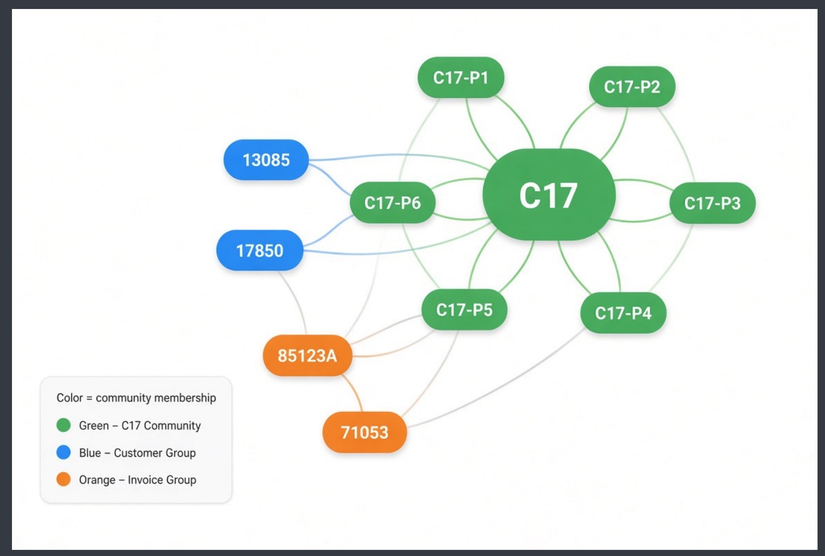
TYPE 1 · PRODUCT COMMUNITIES

Groups of products that frequently co-occur on the same invoices (e.g., C17 = Home Decor, C08 = Bags). Captures latent purchase affinity beyond explicit categories.

TYPE 2 · CUSTOMER COMMUNITIES

Customers linked through shared product communities (SHARES_COMMUNITY). Reveals peer groups with similar buying patterns for look-alike recommendations.

Retail360: GraphRAG for Evidence-Grounded 60-Day Repurchase Prediction



GraphRAG Enhancement Results & Ablation

Model	PR-AUC	Recall@50	Faithfulness	Notes
Original GraphRAG	0.31	0.42	0.78	RFM + invoice paths only
Community-Enhanced	0.39	0.51	0.86	+ community features & multi-hop
Δ (lift)	+0.08	+0.09	+0.08	Primary experimental signal

Ablation Insights

- Removing community features drops PR-AUC back to ~0.31 — confirming the +0.08 lift is attributable to community membership.
- Multi-hop paths through Community nodes improve evidence coverage and raise faithfulness of cited invoiceIds.
- Small sample graph limits ranking stability; full Online Retail II scale is the next critical experiment.
- No GNN used in this iteration (optional); community structure alone delivered the observed gains.

What meaningful communities emerged?

Product communities (e.g., Home Decor C17, Bags C08) capture co-purchase affinity pure RFM misses. Customer communities formed via shared product membership enable peer look-alikes.

Does community detection improve GraphRAG?

Yes. +0.08 PR-AUC, higher Recall@50, and +0.08 faithfulness. Multi-hop retrieval returns concrete invoice evidence that pure text RAG cannot surface.

How do results justify the application?

The lift and improved evidence coverage directly motivate the three HIGH-priority features: Community-Aware Retrieval, Evidence-Grounded Q&A, and Interactive Community Explorer.

Key limitation acknowledged

Sample graph size constrains ranking tests. All claims about scalability and recommendation quality remain provisional until the full dataset is materialised.

9 · HUMAN DESIGN vs AI DESIGN

Dimension	Human Design	AI Design (Agentic)
Community Method	Leiden only	Leiden + GNN embeddings
Retrieval	Template + multi-hop	NL search + multi-hop + captions
Multimodal	Static sketch of viz + bars	Interactive Plotly + auto captions
Features	3 HIGH + 1 MED + 1 FUTURE	3 HIGH original + 2 new HIGH/MED
Explainability	Invoice citations	Citations + visual + NL summary
Risk of complexity	Low	Managed – GNN is additive only

TOOLS USED

- Grok (this conversation) – primary agentic partner
- Reference notebooks from CS5588 package
- Conceptual exploration of Leiden / Louvain / GNN
- Prompt chaining for feature prioritization
- Ready for Antigravity / Claude Code in class

KEY FINDINGS

- Leiden remains strongest for explainable, well-connected communities
- GNN embeddings are best used as additive features, not as a replacement for community IDs
- NL search + auto-captions close the gap between analysts and managers
- Interactive viz is higher priority than speech/video for this user group

KEY PROMPTS (abridged)

"Propose 3 community detection alternatives for a co-purchase graph that must preserve temporal cutoff and invoice provenance."

"Generate multimodal explanation designs that turn the +0.08 community lift into manager-ready charts."

CONCEPTUAL COMMUNITY GRAPH (AI-refined)



Edges: PURCHASED · BELONGS_TO · SHARES_COMMUNITY | GNN embeddings enrich FeatureVector while community IDs stay human-readable

Acknowledgements

We gratefully acknowledge the organizers and sponsors of **Hack-A-Roo 2026** for providing an outstanding platform to explore innovative applications of data science.
Special thanks to the instructors and teaching staff of **CS 5588 – Data Science** at the University of Missouri-Kansas City (UMKC) for their guidance, mentorship, and support throughout this project.
We also thank UMKC for the computational resources and academic environment that made this work possible.